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Course Title: Cloud Computing
Course Code: SWE4002

Digital Assignment - 1

1. How can VIT university implement different types of cluster computing systems such as high-performance computing (HPC) clusters, load balancing clusters and storage clusters to enhance the efficiency and and reliability of its academic research & administrative operations including managing Large-scale student data, supporting computationally intensive research projects and ensuring uninterrupted access to online resources?

Cluster Computing implementation strategy for VIT University:

1. High-Performance Computing (HPC) clusters:

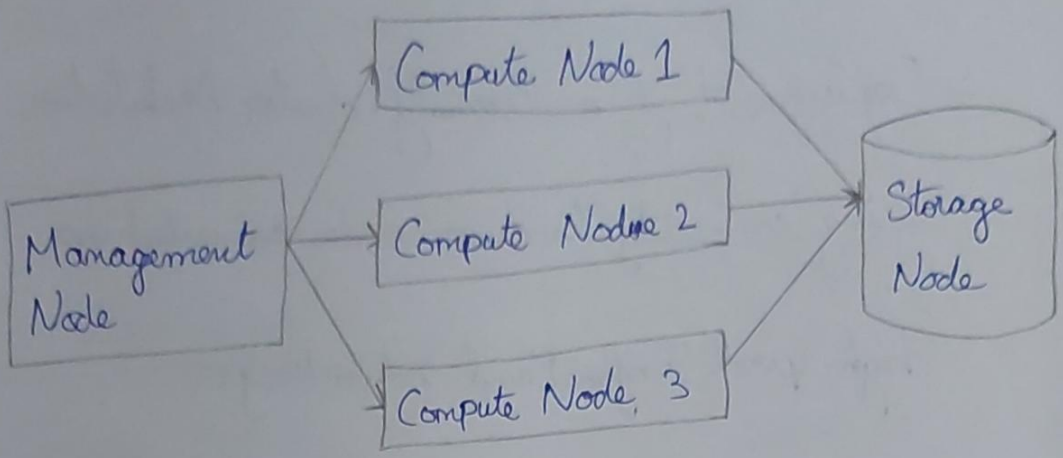
VIT university can implement HPC clusters to support computationally intensive research projects and academic workloads through:

→ Deployment of a Beowulf cluster Architecture using commodity hardware nodes connected via high-speed InfiniBand networking.

→ Installation of parallel computing middleware like OpenMPI for distributed processing.

→ Implementation of job-scheduling systems (eg. SLURM or PBS) to manage research workloads.

→ Dedicated nodes for specific departments (Physics Simulations, Bioinformatics analysis, Engineering modelling).



2. Load-Balancing Clusters:

To ensure uninterrupted access to online resources and applications, VIT can implement:

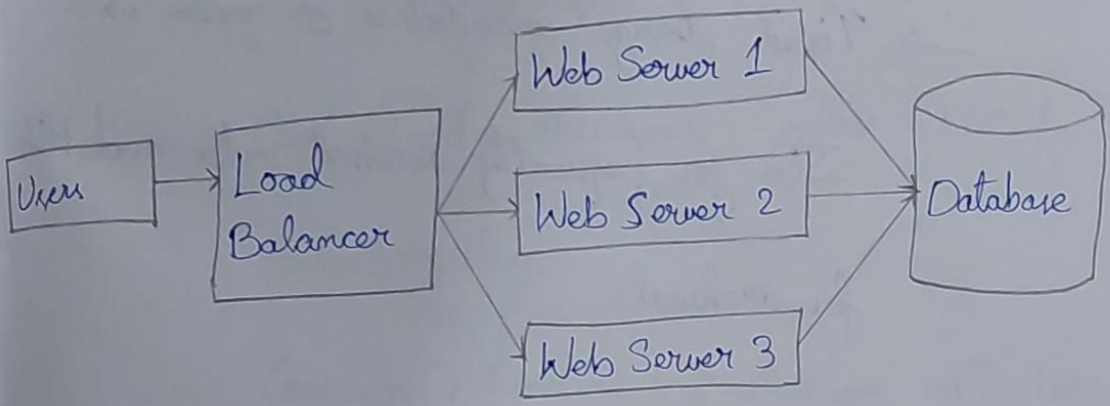
→ Application Delivery Controllers (ADCs) to distribute student portal and LMS traffic across multiple Web Servers.

→ Round-robin DNS with health-checking for even distribution of network loads.

→ Active-active configuration for critical services

with automatic failover.

→ Session persistence for maintaing student login states across different servers.



3. Storage Cluster:

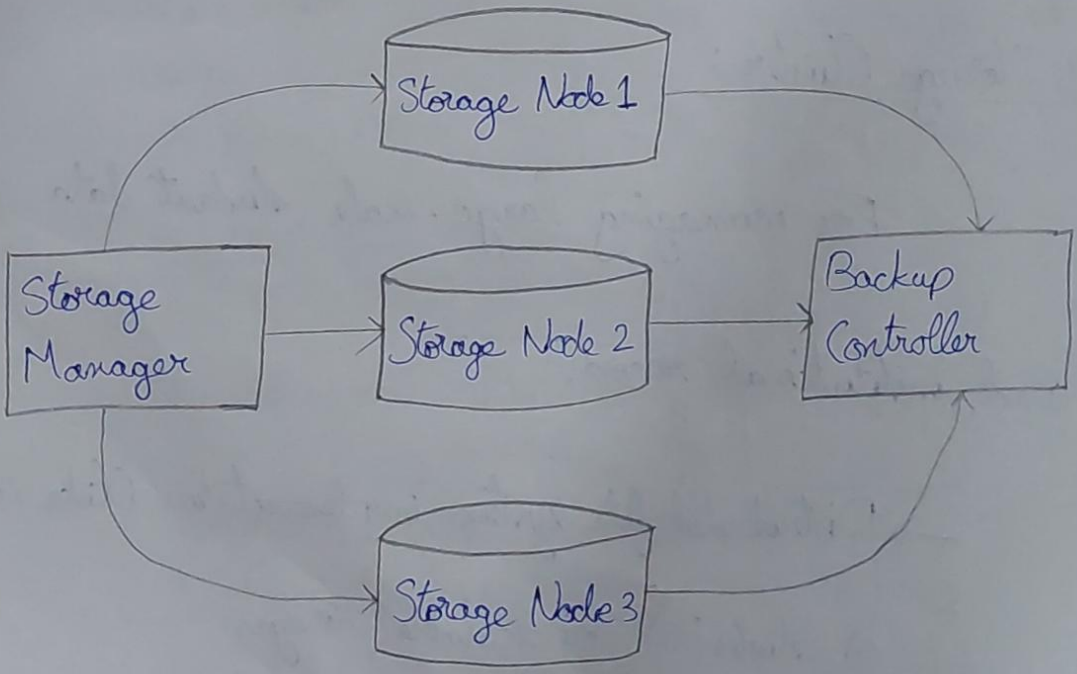
For managing large-scale student data and institutional records:

→ Distributed file system implementation (like Ceph or GlusterFS) for scalable storage.

→ Redundant Array of Independent Nodes (RAIN) architecture for data reliability.

→ Automated backup and disaster recovery systems.

→ Tiered storage architecture or system with SSDs for frequently accessed data and HDDs for archival.



Integration and Management:

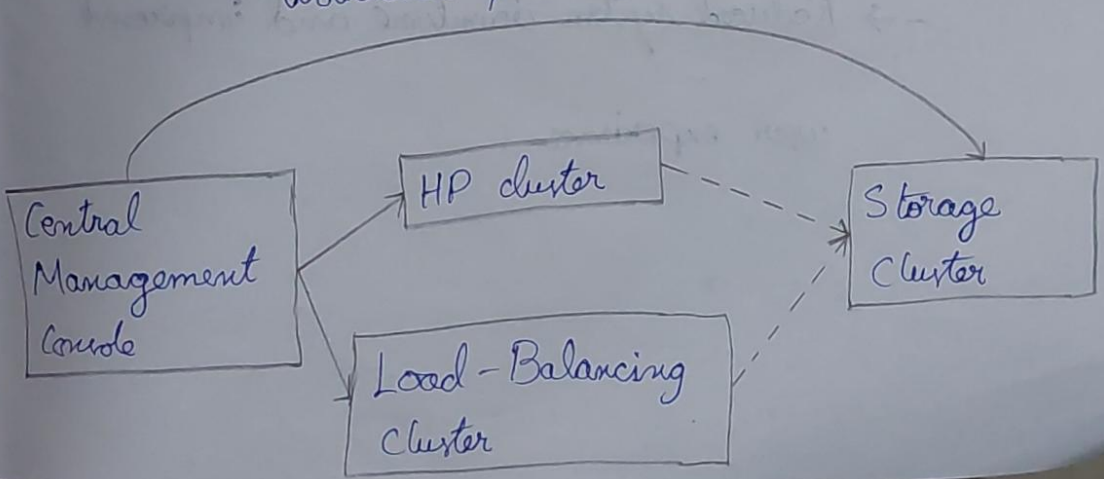
The three cluster types can be Integrated through:

→ Unified management interface for monitoring all cluster resources.

→ Centralized authentication system (LDAP/ Active directory).

→ Common networking infrastructure with VLAN's for security segregation.

→ Automated resource Allocation based on Workload priorities.



Benefits and Outcomes:

This implementation will provide:

→ Enhanced research capabilities through powerful computing resources.

→ Improved system reliability with redundancy and failover.

→ Scalable storage solutions for growing institutional data.

→ Better resource utilization through load Balancing.

→ Reduced system downtime and improved user experience.

Technical Requirements:

Essential components for succesful implementation:

→ High-speed network backbone (minimum 40GBPS).

→ Enterprise-grade servers with redundant power supplies.

→ Storage systems with hot-swappable drivers.

→ Monitoring and management software suite.

→ Regular backup and disaster recovery procedures.

2. How can VIT university leverage different types of cloud computing models - public, private, hybrid and community clouds to optimize its academic, research and administrative operations such as providing scalable e-learning platforms, supporting research collaborations, managing sensitive student and staff data securely and ensuring cost-effective infrastructure for university-wide digital transformation.

Cloud Computing implementation strategy for VIT university

1. Public cloud implementation:

VIT can leverage public cloud services for

→ E-learning platforms using AWS or Azure's scalable infrastructure.

→ Learning Management Systems (LMS) hosting with auto-scaling capabilities.

→ Student collaboration tools and email services through Google Workspace or Microsoft 365.

→ Public-facing websites and admission portals with global content delivery

→ Development and Testing environments for student projects.

2 Private cloud deployment :

For sensitive operations and Data, VIT should maintain:

→ On-premises open stack or VM ware-based private cloud.

→ Secure student information systems and academic records.

→ Financial management and Payroll systems.

→ Internal research data and Intellectual property.

→ Administrative applications requiring strict compliance.

→ High-performance computing resource for specialized research.

3. Hybrid Cloud Architecture:

Integration of public and private clouds through:

→ Unified Identity management across all platforms.

→ Secure VPN connections between on-premises and public cloud services.

→ Burst capability for handling peak loads during admission.

→ Disaster Recovery and backup solutions.

→ Resource allocation based on workload sensitivity and performance requirements.

b. Community cloud solutions:

Collaboration with other educational institutions through:

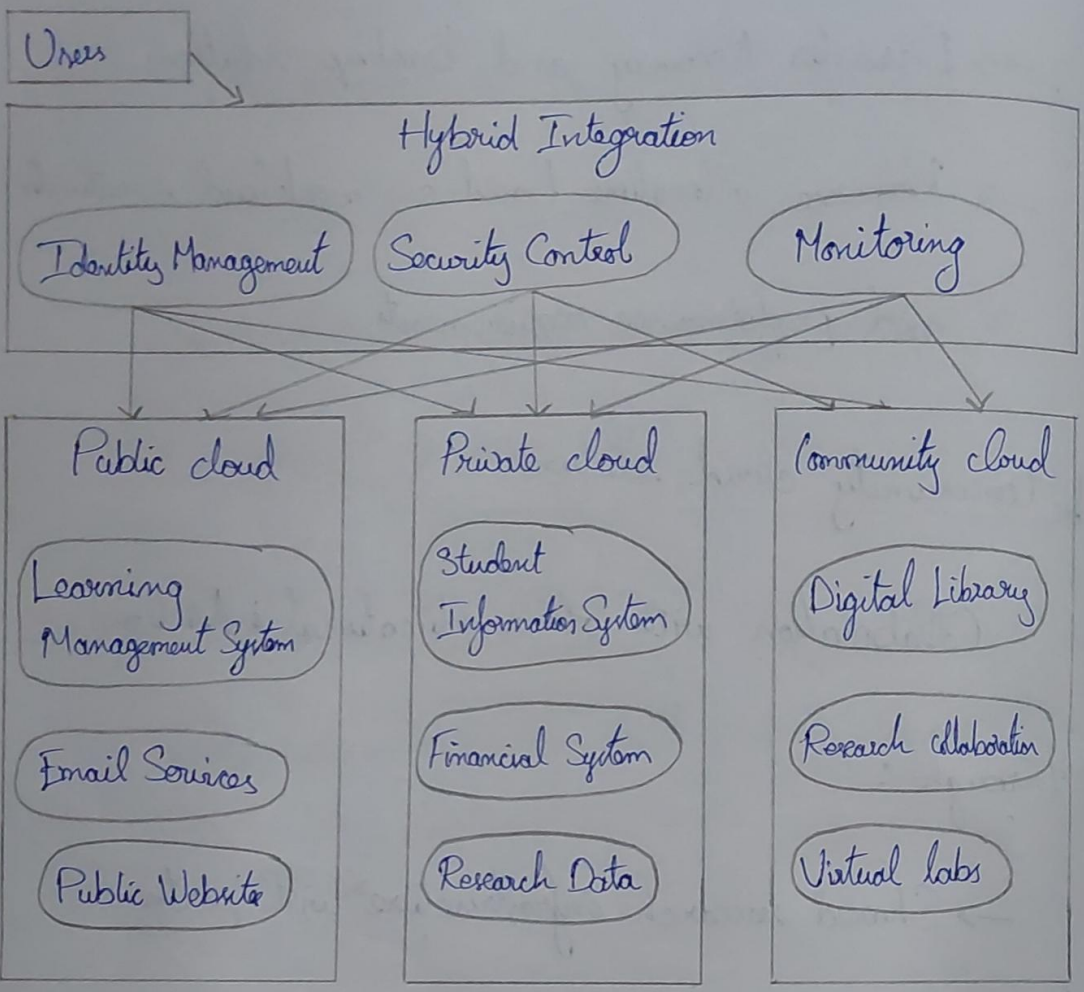
→ Shared research infrastructure with partner universities.

→ Common digital library resources.

→ Joint virtual laboratories.

→ Shared High-Performance Computing facilities.

→ Collaborative research data repositories.



Implementation Strategy:

Security and Compliance:

→ Multifactor authentication for all cloud services.

→ Data encryption at rest and in transition.

→ Regular Security Audits and compliance checks.

→ Role-based access control (RBAC)

→ Data residency compliance for student records.

Cost Optimization:

→ Resource usage monitoring and optimization

→ Auto-Scaling policies for demand management.

→ Reserved instance for predictable workloads.

→ Pay-as-you-go model for variable demands.

→ Regular cost analysis and Optimization reviews.

Benefits:

→ Enhanced scalability for educational services.

→ Improved collaboration capabilities.

→ Better resource utilization.

→ Reduced infrastructure maintenance overhead.

→ Increased reliability and availability.

→ Flexible access to computing resource.

Migration and Integration plan:

→ Phased migration approach.

→ Integration with existing systems.

→ Training programs for staff and students.

→ Regular performance monitoring.

→ Continuous improvement process.